Prarthana Sewmini

Associate AI Engineer

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LinkedIn GitHub Website

PROFILE

Engineering the intelligence behind what users see. Working at the intersection of data, machine learning, and real world systems to turn raw information into reliable AI driven decisions. Focused on building models and pipelines that quietly do the heavy lifting so products feel simple, fast, and smart.

PROFESSIONAL EXPERIENCE

Associate AI Engineer, *Vectorium Labs Pvt Ltd* 01/2026 – Present

- Architected and implemented a production-ready RAG system, handling data ingestion, embedding, retrieval, and response generation.
- Owned delivery of AI capability upgrades, from design to deployment, under minimal supervision.

AI Engineer Intern, *Vectorium Labs Pvt Ltd* 07/2025 – 01/2026

- Designed and developed AI-driven solutions using Python, focusing on machine learning and data-driven decision systems.
- Built and optimized data pipelines for preprocessing, feature engineering, and model evaluation.
- Collaborated with cross-functional teams to translate business requirements into scalable AI solutions.

AI Developer Intern, *StrataNEXTGEN PTY LTD* 06/2025 – 09/2025

- Built and integrated AI-powered chatbot features with API connections and automated PDF generation.
- Developed role-based dashboards and interactive UIs using Next.js, TypeScript, and Tailwind CSS.
- Collaborated on API contracts, Supabase database schema, and deployments on AWS Amplify & Google Cloud Run.

SKILLS

- Python
- Prompt Engineering & LLM Optimization
- FastAPI, REST API Development
- Git & GitHub
- LangChain (AI workflow integration)
- RAG
- Generative AI

EDUCATION

General Sir John Kotelawala Defence University, 2022 – 2025
BSc (Hons) Data Science & Business Analytics Ratmalana, Sri Lanka

Anula Vidyalaya, Nugegoda, 2011 – 2020
GCE Advanced Level Examination - Physical Science Stream Nugegoda, Sri Lanka

PROJECTS

Client Project - AI-powered Weather Application
Jul 2025 - Dec 2025

- Designed and implemented a RAG pipeline combining weather API data and embedded knowledge sources for enhanced AI responses
- Built end-to-end data flow including ingestion, preprocessing, retrieval, and LLM-based generation
- Owned AI system upgrades from implementation through testing and deployment

Technologies used: Python, LLMs, FastAPI, AWS, Hugging Face, Runpod

Face Recognition Attendance System
April 2025

- Designed and implemented a real-time facial recognition system using Python, DeepFace (FaceNet), and MTCNN to automate attendance tracking for university lecture slots (8:00-11:00, 11:30-14:30).
- Developed a Flask-based web interface with manual attendance logging, latecomer alerts (audio-visual feedback post-9:30 AM), and CSV export functionality, enhancing administrative efficiency.
- Improved recognition accuracy by expanding the dataset.

Technologies used: Python, OpenCV, Pandas, Flask, HTML/CSS/JavaScript

Kidney Disease Classification with Deep Learning
March 2025

- Developed a deep learning model using Python, TensorFlow, and Keras to classify kidney disease from medical imaging, support ing early diagnosis.
- Implemented data preprocessing with OpenCV and Pandas, model training, and experiment tracking with MLflow to enhance classification accuracy.
- Demonstrated proficiency in computer vision, model optimization, and version control, with code hosted on GitHub for reproducibility.

Technologies used: Python, TensorFlow, Keras, OpenCV, Pandas, MLflow, Flask

CERTIFICATES

- AI Agents Fundamentals - Huggingface
- Artificial Intelligence Fundamentals (IBM)
- Machine Learning for Data Science Projects (IBM)
- Generative AI Foundations (Databricks)
- AWS Educate Introduction to Generative AI
- AWS Educate Machine Learning Foundations

PERSONAL SKILLS

- Technical Leadership
- AI System Design
- Model Evaluation
- Production Deployment
- Performance Optimization
- Cross-functional Collaboration
- Technical Communication
- Consistency & Discipline

REFERENCES

Dr. (Maj) RMM Pradeep (Retd), *Senior Lecturer Grade I,*
Department of Information Technology, Faculty of Computing, General Sir John Kotelawala Defence University
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